Football Yard Booking Website

We want to build a website for booking football yards. There will be two interfaces: one for yard owners and the other for users.

The first interface, designed for yard owners, will feature a scheduling system to manage bookings. Upon signing in, yard owners will need to input all relevant information about their yard, including its size and any additional services they offer, such as lockers and seating for fans.

The second interface is for users and is divided into two scenarios.

1. **Case One**

: If a user has a team and wants to book a yard, the system will detect their location and suggest all nearby yards along with their availability schedules.

2. **Case Two**

: If a user does not have a team, they can select a date for when they want to play and look for players to form a team. This can be done by sending invitations to teams that have not filled their spots for that day or by reaching out to other users in the same area. They can also check for available yards at their desired location.

In both cases, users must sign in and provide their information, which includes their ID number, a photo of their ID, a personal photo, and, if applicable, photos of their yard. They will also need to provide their date of birth and phone number.

Identity verification is required for both interfaces, and user locations must be within a 2 km radius of the yard. Users can book the yard they want through an offer page that displays all yards and their schedules. During the booking process, payment will be made on-site, and the funds will be transferred immediately to the yard owner's bank account. Each day, the system will calculate how much each yard has been booked, and the earnings will be transferred to the yard owner at the end of the day.

System Flow Diagram

```
User/Yard Owner * Sign Up & Verify Identity * Access Dashboard
      |
      |-- (If Yard Owner) * Add Yard Details * Manage Schedule & Bookings
* Track Earnings
      |
      |-- (If Player)
            |
            |-- Has a Team * Find Nearby Yards * View Schedule * Book & Pa
y
            |
            |-- No Team * Select Date * Find Teams/Players * Check Yard Ava
ilability * Book & Pay
```

Key Entities and Attributes

1. **User**
   - UserID (Primary Key)
   - Name
   - IDNumber
   - IDPhoto
   - PersonPhoto
   - DateOfBirth
   - PhoneNumber
   - Email
   - Password
   - Role (Yard Owner / Regular User)
   - Location (for proximity-based suggestions)

2. **Yard Owner**
    - UserID (Foreign Key referencing User)
    - YardID (Primary Key for Yard)
    - YardName
    - YardLocation
    - YardArea
    - ServicesOffered (e.g., lockers, stands, etc.)
    - YardPhotos
    - Schedule (availability slots)
3. **Yard**
    - YardID (Primary Key)
    - YardName
    - YardLocation
    - YardArea
    - ServicesOffered
    - YardPhotos
    - OwnerID (Foreign Key referencing Yard Owner)
4. **Booking**
    - BookingID (Primary Key)
    - UserID (Foreign Key referencing User)
    - YardID (Foreign Key referencing Yard)
    - BookingDate
    - BookingTime
    - PaymentStatus
    - AmountPaid
    - PaymentMethod

5. **Team**
    - TeamID (Primary Key)
    - TeamName
    - CaptainID (Foreign Key referencing User)
    - Members (list of UserIDs)
6. **Offer**
    - OfferID (Primary Key)
    - YardID (Foreign Key referencing Yard)
    - OfferDetails
    - ValidUntil
7. **Payment**
    - PaymentID (Primary Key)
    - BookingID (Foreign Key referencing Booking)
    - Amount
    - PaymentDate
    - PaymentMethod
    - Status (Completed/Pending)
8. **Location**
    - LocationID (Primary Key)
    - YardID (Foreign Key referencing Yard)
    - UserID (Foreign Key referencing User)
    - Proximity (within 2km)

Relationships

1. **User** can be either a **Yard Owner** or a **Regular User**.
2. **Yard Owner** owns one or more **Yards**.

3. **Yard** has a **Schedule** and can have multiple **Bookings**.
4. **Booking** is made by a **User** for a specific **Yard**.
5. **Team** is created by a **User** and can book a **Yard**.
6. **Offer** is associated with a **Yard** and displayed to **Users**.
7. **Payment** is linked to a **Booking**.
8. **Location** is used to filter **Yards** within 2km of the **User**.

ER Diagram

Below is a textual representation of the ER diagram. You can use tools like Lucidchart, Draw.io, or any ER diagram tool to visualize it.

```
⌫User] ---(1)---›  owns ⌃ ---(0..*)--- ⌫Yard Owner]
⌫User] ---(1)---›  creates ⌃ ---(0..*)--- ⌫Team]
⌫User] ---(1)---›  books ⌃ ---(0..*)--- ⌫Booking]
⌫Yard Owner] ---(1)---›  manages ⌃ ---(1..*)--- ⌫Yard]
⌫Yard] ---(1)---›  has ⌃ ---(1..*)--- ⌫Schedule]
⌫Yard] ---(1)---›  offers ⌃ ---(0..*)--- ⌫Offer]
⌫Booking] ---(1)---›  linked to ⌃ ---(1)--- ⌫Payment]
⌫Booking] ---(1)---›  for ⌃ ---(1)--- ⌫Yard]
⌫Team] ---(1)---›  books ⌃ ---(0..*)--- ⌫Yard]
⌫Location] ---(1)---›  filters ⌃ ---(1..*)--- ⌫Yard]
⌫Location] ---(1)---›  filters ⌃ ---(1..*)--- ⌫User]
```

Scenario Cases

1. **Yard Owner Registration**
   - Yard Owner signs up, provides personal details, and adds yard information (location, area, services, photos, etc.).
   - System verifies identity and approves the account.
2. **User Registration**

- User signs up, provides personal details, and uploads required documents (ID, photo, etc.).
- System verifies identity and approves the account.

3. **Yard Booking (User with Team)**
   - User selects "Book a Yard" and provides location.
   - System suggests yards within 2km and displays their schedules.
   - User selects a yard and time slot, confirms booking, and makes payment.
4. **Yard Booking (User without Team)**
   - User selects "Find a Team" and provides a date.
   - System suggests available teams or allows the user to create a new team.
   - Once a team is formed, the user books a yard as above.
5. **Offer Display**
   - System displays all yards with their schedules and any ongoing offers.
   - Users can filter by location, date, and services.
6. **Payment Processing**
   - User pays for the booking via the website.
   - Payment is transferred to the yard owner's account immediately.
   - Daily earnings are calculated and transferred to the yard owner at the end of the day.
7. **Identity Verification**
   - Both users and yard owners must verify their identity during registration.
   - System checks ID photos and personal photos against provided details.
8. **Proximity Filtering**
   - System ensures that only yards within 2km of the user's location are suggested.